If we define the ratio

$$\alpha = \sigma_\gamma/\sigma_f = \Sigma_\gamma/\Sigma_f \tag{19.5}$$

(cf. (11.6)) the probability that the captured neutron gives rise to fission is $\sigma_f/(\sigma_f + \sigma_\gamma) = 1/(1 + \alpha)$. The number of neutrons produced for each neutron captured is

$$\eta = \nu/(1 + \alpha) \tag{19.6}$$

where η is the *neutron yield per absorption*, also called the *fission factor*. A primary requirement for a chain reaction is that $\eta > 1$. Table 19.2 contains η-values for the most important nuclides. At thermal energies η is highest for ^{233}U, while for fast neutrons η is highest for ^{239}Pu.

In mixtures of isotopes the macroscopic cross section Σ must be used in calculating α. Since for natural uranium $\Sigma_f = 0.72 \times 582$ per 100 and $\Sigma_\gamma = (0.72 \times 99 + 99.3 \times 2.70)$ per 100, one obtains $\alpha = 0.81$ and $\eta = 1.34$, which means a chain reaction is possible with thermal neutrons in natural uranium.

19.5. NEUTRON MODERATION

The mode of moderation of the neutrons is one of the crucial design features of a thermal reactor. The fast fission neutrons lose their kinetic energy through elastic scattering with the atoms of the moderator and construction material. In §10.5 the equations are given for the energy change in such collision processes, but only for head-on collisions. Because most collisions involve angular scattering, the number of collisions required to reduce a fast neutron to thermal energy is larger. The *average logarithmic energy decrement* is given by

$$\xi = \overline{\ln\left(E_n/E_n'\right)} = 1 + \left[(A-1)^2/(2A)\right]\ln[(A-1)/(A+1)] \tag{19.7}$$

where E_n' is the neutron energy after collision with a moderator atom of mass number A. From Table 19.3 it is seen that "light" water (i.e. H_2O) is most effective in reducing the

Table 19.3 Physical constants of some moderator materials

Property	H_2O	D_2O	Be	C
$N_0 \times 10^{-30}$ (atoms m^{-3})	0.0333	0.0332	0.123	0.0837
ρ (kg m^{-3})	997	1104	1848	1670
σ_a(th) (barns)	0.66	0.0013	0.0076	0.0035
Σ_a(th) (m^{-1})	2.2	0.0044	0.094	0.029
σ_s(epith) (barns)	49	10.6	5.9	4.7
ξ	0.927	0.510	0.206	0.158
$\xi \times \Sigma_s$(epith)/Σ_a(th)	68	4095	160	211
L_m^2(m^2)	0.000713	1.51	0.043	0.287
τ (m^2)	0.0031	0.0125	0.0097	0.0349

neutron velocity. The average number of collisions n required to reduce the neutron energy from E_n^0 to E_n is given by

$$n = \xi^{-1} \ln\left(E_n^0/E_n\right) + 1 \tag{19.8}$$

The *slowing down power* (*SDP*) of a moderator depends in addition on the neutron scattering cross section and number of scattering atoms per unit volume (N_0):

$$SDP = \xi\, N_0\, \sigma_s = \xi\, \Sigma_s \tag{19.9}$$

The *SDP* is an average value over the epithermal neutron energy region. A good neutron moderator should divert few neutrons from the fission process, i.e. the neutron absorption cross section must be small. In this respect both heavy water (i.e. D_2O) and carbon are superior to H_2O, since for H_2O the reaction probability for $^1H(n,\gamma)^2H$ is relatively large (Table 19.3). In order to incorporate this property, the concept *moderating ratio* (*MR*) is used as a criterion of the moderator property in the thermal neutron energy region:

$$MR = SDP/\Sigma_a = \xi\, \Sigma_s/\Sigma_a \tag{19.10}$$

The moderating properties are summarized in Table 19.3. The moderator qualities decrease in the order $D_2O > C > Be > H_2O$. In a commercial reactor, price and other properties must also be considered, so H_2O is favoured over D_2O in most reactors.

19.6. THE NEUTRON CYCLE

In order for a chain reaction to occur at least one of the neutrons released in fission must produce a new fission event. This condition is defined by the *multiplication factor k*:

$$k = \frac{\text{number of neutrons in generation 2}}{\text{number of neutrons in generation 1}} \tag{19.11}$$

when $k > 1$ the number of neutrons in the second generation exceeds the number of neutrons consumed. Under this condition, the neutron flux, and, thus, the number of fission events increase for each successive neutron generation with a resultant increase in the power production of the reactor (§19.8 discusses further the *reactor kinetics*). If $k = 1$ the number of fissions per unit time, and thus the energy production, is constant. For $k < 1$ the chain reaction cannot be maintained. A reactor operating with $k = 1$ is said to be *critical*, $k > 1$ *supercritical*, and $k < 1$ *subcritical*. k is regulated by means of the control rods and in PWRs also by the n-absorber concentration in the water (usually boric acid).

In any neutron generation the neutrons experience a variety of fates (Fig. 19.6). Some neutrons escape the reactor and some are absorbed in the reactor structural materials and shielding (i.e. fuel cans, control rods, moderator, coolant, etc.). To take this loss into